

Prelim-Code-P3.R

stein

2026-04-19

```
library(survminer)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: ggpubr
```

```
library(survival)
```

```
##
```

```
## Attaching package: 'survival'
```

```
## The following object is masked from 'package:survminer':
```

```
##
```

```
##      myeloma
```

```
library(gtsummary)
```

```
library(MASS)
```

```
##
```

```
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:gtsummary':
```

```
##
```

```
##      select
```

```
# Upload Data set
```

```
fhsdata <- read.csv("C:/Users/stein/OneDrive/Documents/School/2026 Spring/Advanced Methods/BIOS6624/Pro
```

```
# Notes to self:
```

```
# remove stroke = 1 at baseline
```

```
# remove stroke >3652.5 days
```

```
# check missing: BP(systolic), age, diabetes
```

```
# heart disease, smoking status, bp medication, HDL/LDL, BMI
```

```
#
```

```
## Initial data cleaning
```

```

# remove anyone with stroke at baseline (STROKE = 1 at period 1)
clean <- fhsdata[!(fhsdata$TIMESTRK == 0 & fhsdata$PERIOD == 1), ]

# filter to use just the baseline values
base <- clean[clean$PERIOD == 1, ]

# censor follow-up to 10 years (365.25*10)
base$stroke_event <- ifelse(base$TIMESTRK <= 3652.5 & base$STROKE == 1, 1, 0)
base$stroke_time <- ifelse(base$TIMESTRK <= 3652.5, base$TIMESTRK, 3652.5)

# check covariates for missing
sum(is.na(base$SYSBP)) # 0

```

```
## [1] 0
```

```
sum(is.na(base$AGE)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$DIABETES)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$CURSMOKE)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$BMI)) # 17
```

```
## [1] 17
```

```
sum(is.na(base$TOTCHOL)) # 52
```

```
## [1] 52
```

```
sum(is.na(base$BPMEDS)) # 60
```

```
## [1] 60
```

```
sum(is.na(base$HEARTRTE)) # 0
```

```
## [1] 1
```

```

# check covariates for implausible values
sum(base$SYSBP > 300 | base$SYSBP < 60) # 0

```

```
## [1] 0
```

```

sum(base$AGE > 99 | base$AGE <= 18) # 0

## [1] 0

min(base$BMI, na.rm = TRUE) # 15.54

## [1] 15.54

max(base$BMI, na.rm = TRUE) # 56.8 all plausible values for BMI

## [1] 56.8

### fit KM curves to determine what we put in the cox model

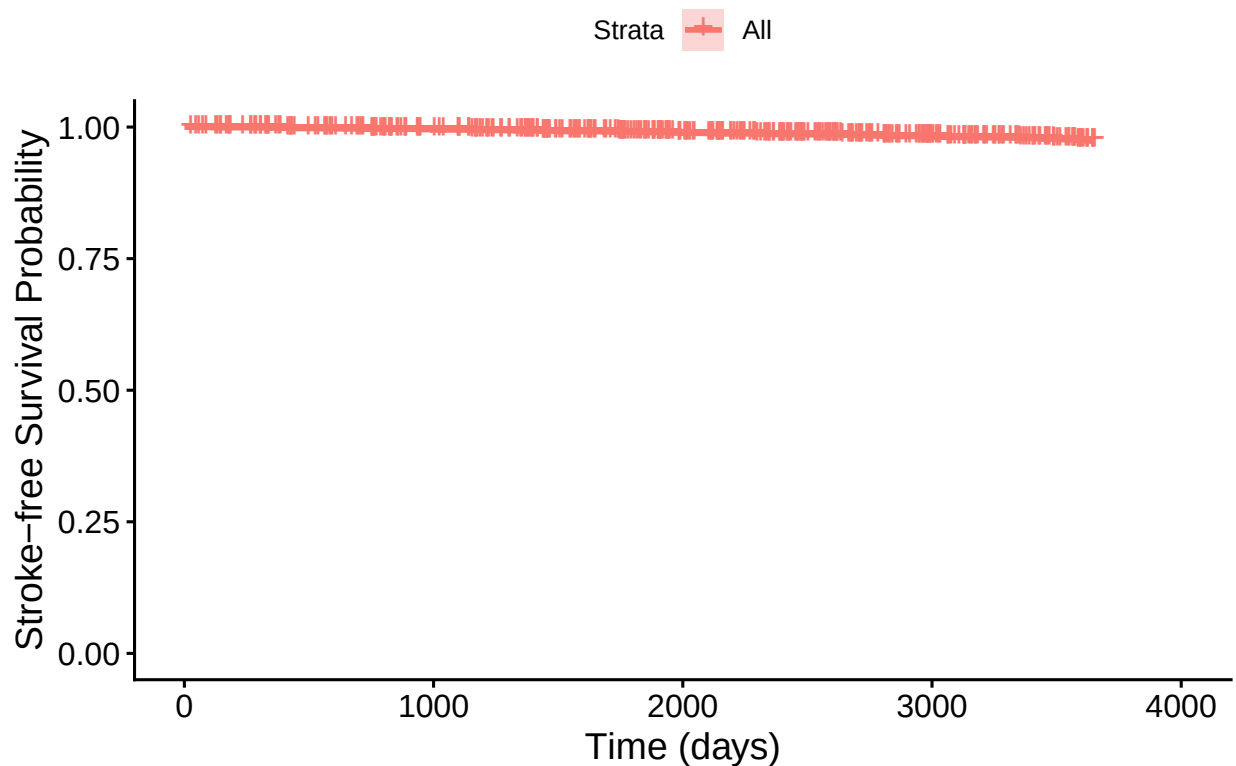
surv_obj <- Surv(time = base$stroke_time, event = base$stroke_event)

# not stratified
km_overall <- survfit(surv_obj ~ 1, data = base)

ggsurvplot(km_overall,
  data = base,
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "Overall Kaplan-Meier Curve")

```

Overall Kaplan-Meier Curve

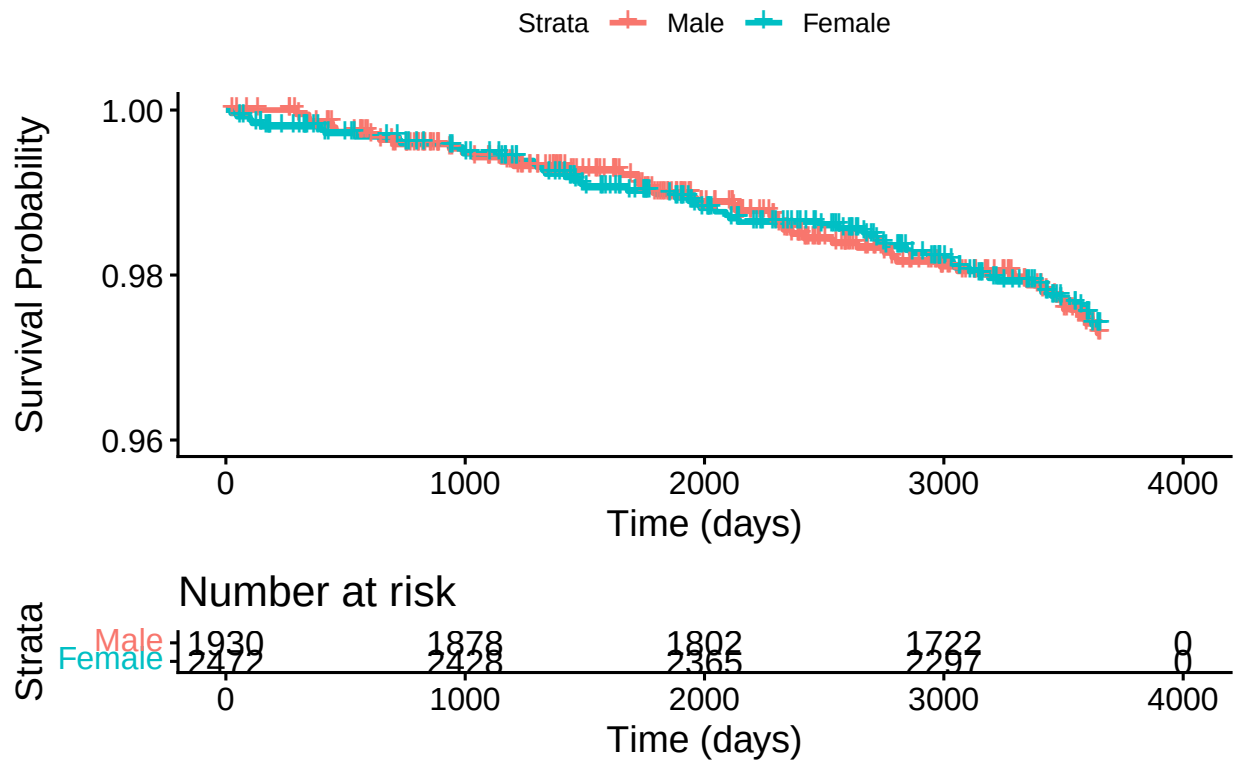


```
# stratified by sex
km_sex <- survfit(surv_obj ~ SEX, data = base)

ggsurvplot(km_sex,
  data = base,
  legend.labs = c("Male", "Female"),
  xlab = "Time (days)",
  ylab = "Survival Probability",
  title = "Kaplan-Meier by Sex",
  ylim = c(0.96, 1),
  break.y.by = 0.02,
  risk.table = T)
```

```
## Ignoring unknown labels:
## * colour : "Strata"
```

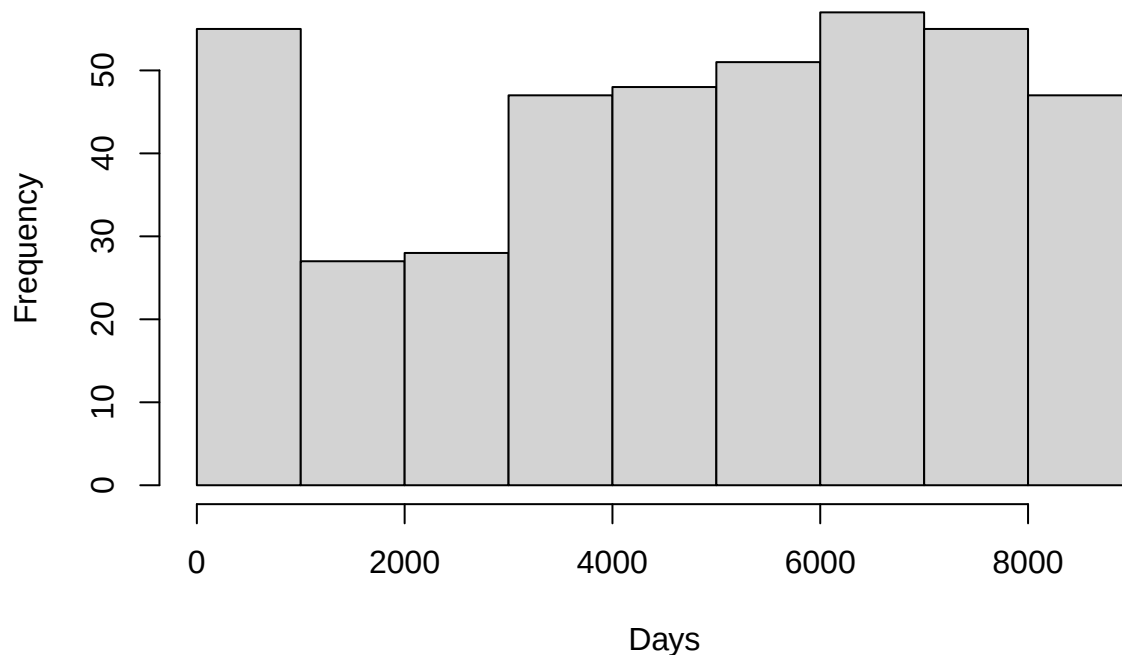
Kaplan-Meier by Sex



```
##### FILTERED DATA WRONG ##### fix it

hist(fhsdata$TIMESTRK[fhsdata$PERIOD == 1 & fhsdata$STROKE == 1],
  main = "TIMESTRK for Stroke=1 at Period 1",
  xlab = "Days")
```

TIMESTRK for Stroke=1 at Period 1



```
summary(fhsdata$TIMESTRK[fhsdata$PERIOD == 1 & fhsdata$STROKE == 1])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##         0   2769    5134   4740   6952   8702
```

```
### check the variables that we are debating including
```

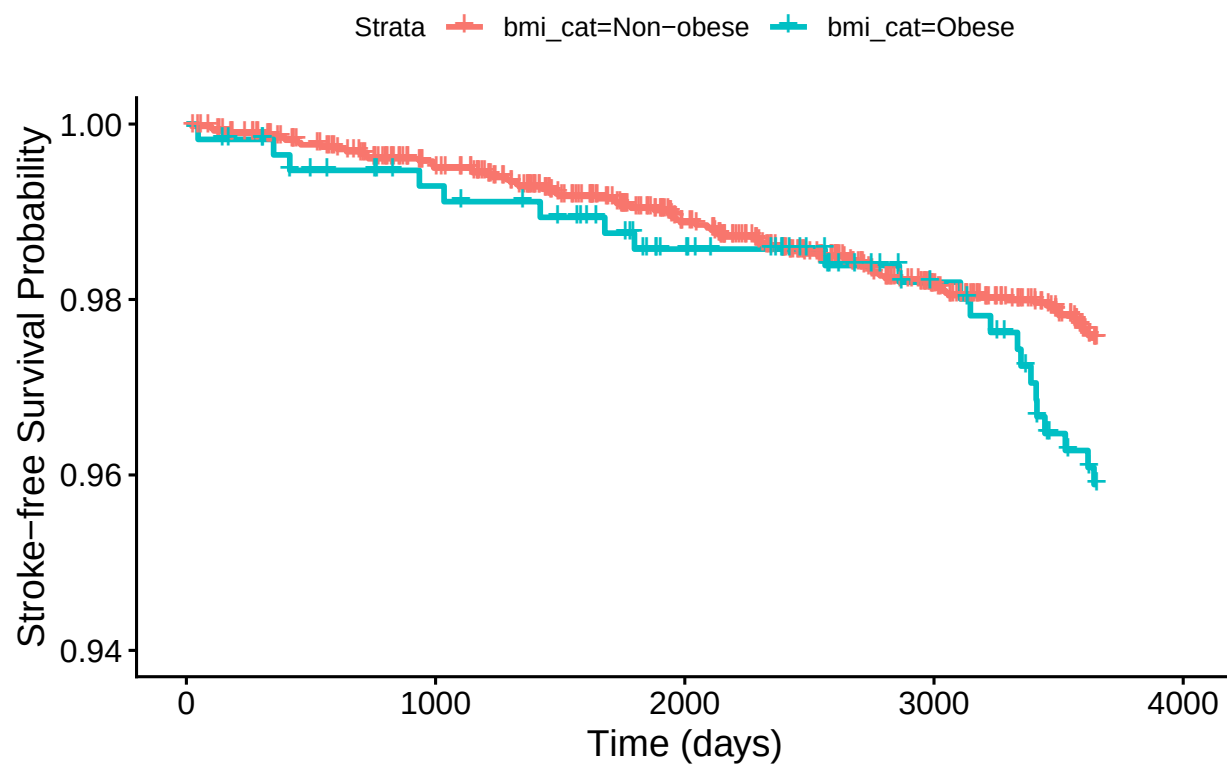
```
# Dichotomized versions are just for KM plotting
```

```
base$bmi_cat    <- ifelse(base$BMI >= 30, "Obese", "Non-obese")
base$chol_cat   <- ifelse(base$TOTCHOL >= 240, "High Chol", "Normal Chol")
base$bpmeds_cat <- ifelse(base$BPMEDS == 1, "On BP Meds", "Not on BP Meds")
base$hearttrte_cat <- ifelse(base$HEARTRTE > 100, "High HR", "Normal HR")
base$smoke_cat  <- ifelse(base$CURSMOKE == 1, "Smoker", "Non-smoker")
```

```
# BMI
```

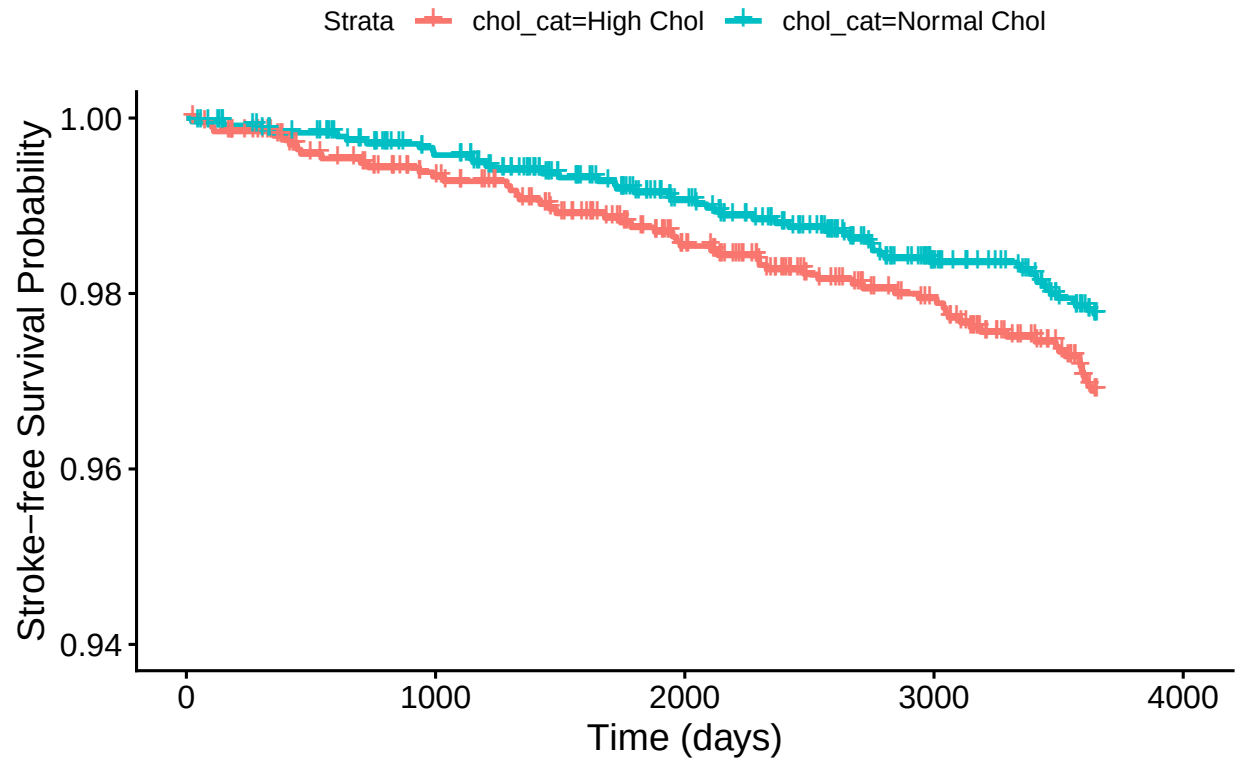
```
km_bmi <- survfit(surv_obj ~ bmi_cat, data = base)
ggsurvplot(km_bmi, data = base,,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by BMI")
```

KM Curve by BMI



```
# Total cholesterol
km_chol <- survfit(surv_obj ~ chol_cat, data = base)
ggsurvplot(km_chol, data = base,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Total Cholesterol")
```

KM Curve by Total Cholesterol



```
# BP meds
km_bpmeds <- survfit(surv_obj ~ bpmeds_cat, data = base)
ggsurvplot(km_bpmeds, data = base,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by BP Meds")
```

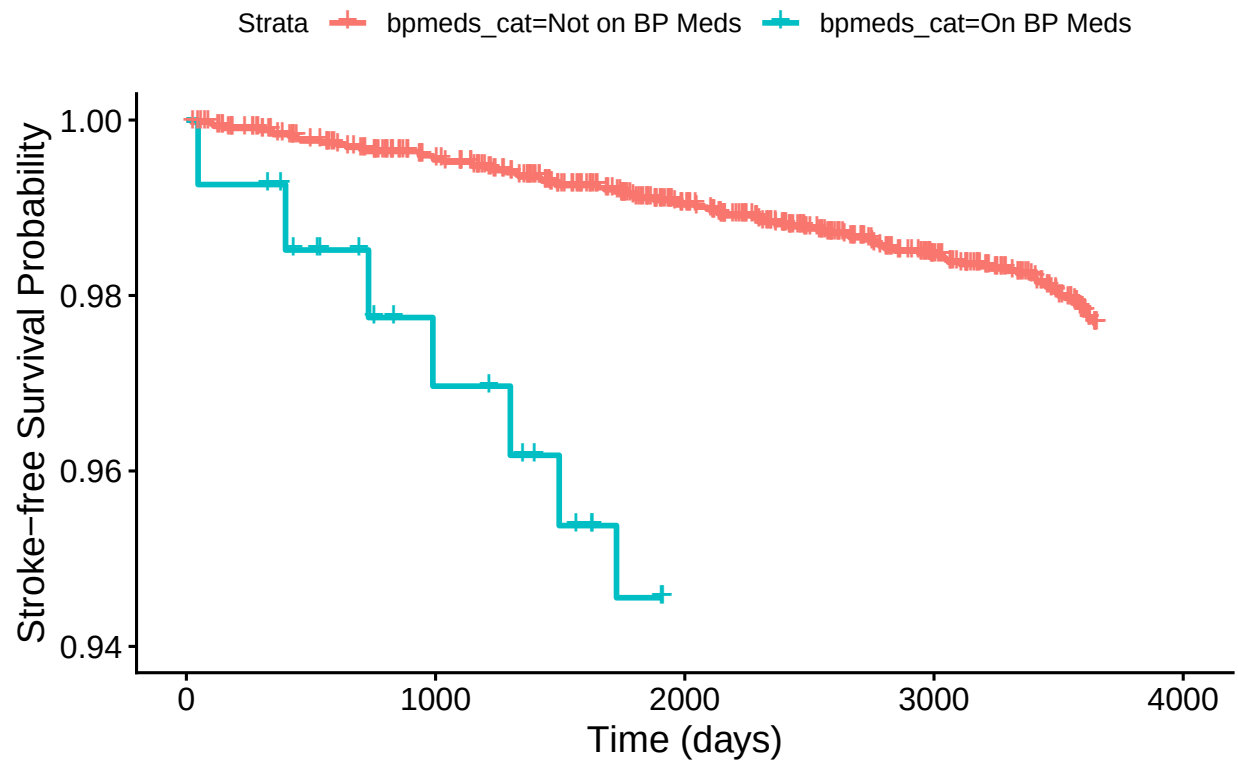
```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_step()').
```

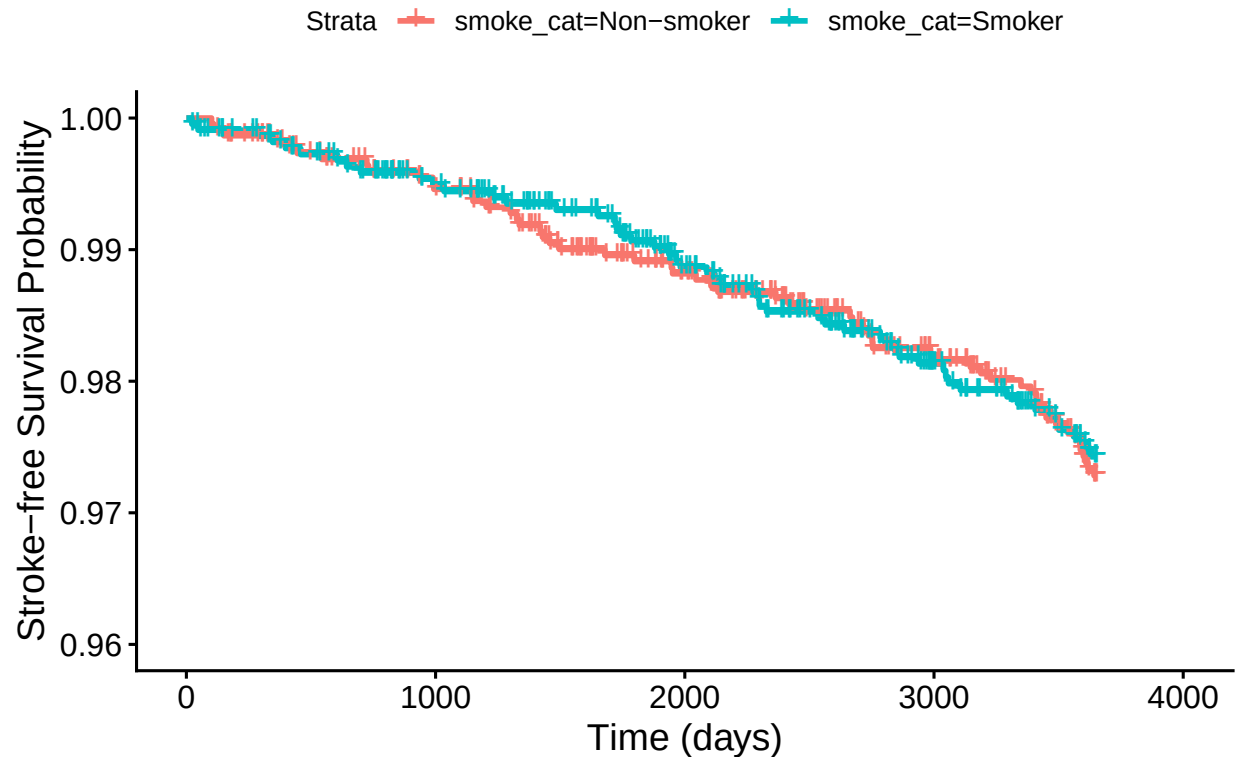
```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```

KM Curve by BP Meds



```
# Smoking
km_smoke <- survfit(surv_obj ~ smoke_cat, data = base)
ggsurvplot(km_smoke, data = base,
  ylim = c(0.96, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Smoking Status")
```

KM Curve by Smoking Status



```
# Heart rate
km_hearttrte <- survfit(surv_obj ~ hearttrte_cat, data = base)
ggsurvplot(km_hearttrte, data = base,
  ylim = c(0.96, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Heart Rate")
```

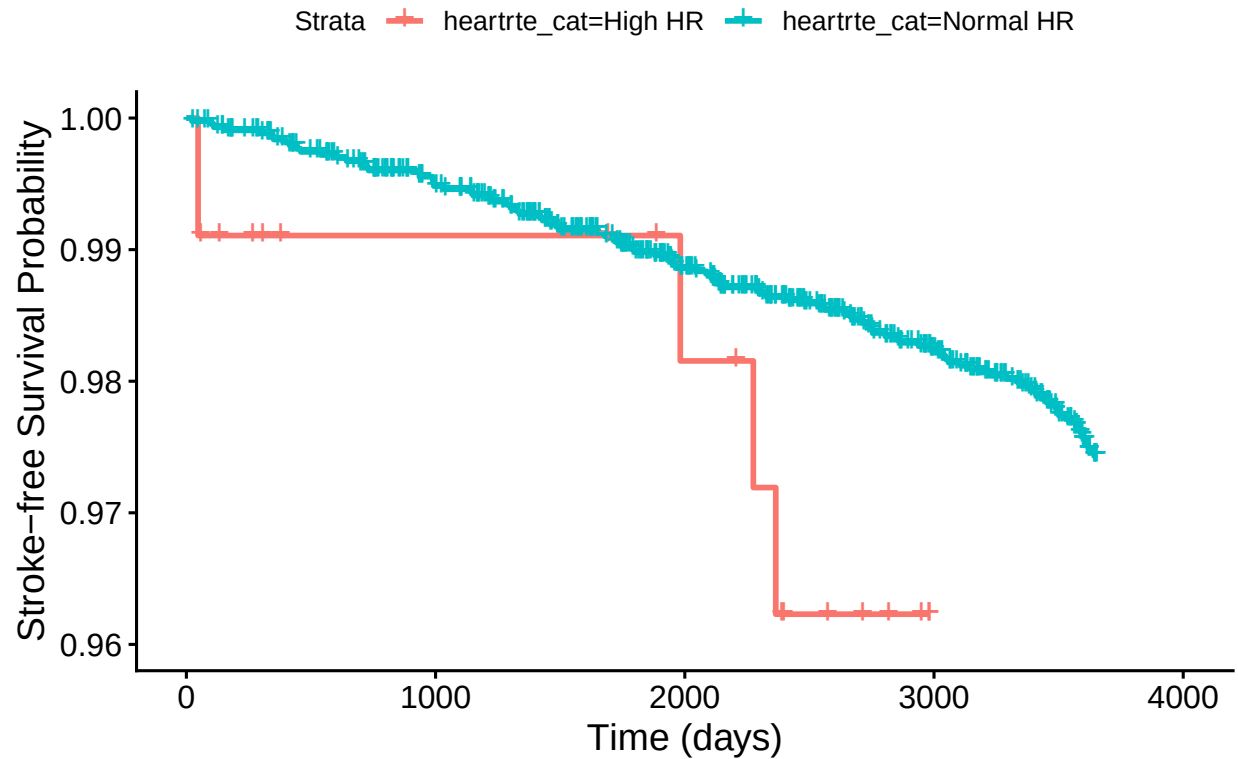
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

KM Curve by Heart Rate



```
### Start survival models ###
```

```
### complete case analysis to add back in dropped variables
```

```
base_final <- base[complete.cases(base[, c("SYSBP", "AGE",
                                             "DIABETES", "BPMEDS"))], ]
nrow(base_final)
```

```
## [1] 4342
```

```
sapply(base[, c("SYSBP", "AGE",
                 "DIABETES", "BPMEDS")], function(x) sum(is.na(x)))
```

```
##      SYSBP      AGE DIABETES      BPMEDS
##         0         0         0         60
```

```
## split into male and female
```

```
base_male <- base_final[base_final$SEX == 1, ]
base_female <- base_final[base_final$SEX == 2, ]
```

```
## update survival objects for both
```

```

surv_male <- Surv(time = base_male$stroke_time,
                  event = base_male$stroke_event)
surv_female <- Surv(time = base_female$stroke_time,
                   event = base_female$stroke_event)

## Full model then stepwise model
# Male
library(MASS)

# Male full model
full_male <- coxph(surv_male ~ SYSBP + AGE + DIABETES + BPMEDS,
                  data = base_male)

# Stepwise on male model
step_male <- stepAIC(full_male,
                    scope = list(lower = ~ SYSBP + AGE + DIABETES,
                                upper = ~ SYSBP + AGE + DIABETES + BPMEDS),
                    direction = "both",
                    trace = TRUE)

```

```

## Start:  AIC=661.41
## surv_male ~ SYSBP + AGE + DIABETES + BPMEDS
##
##           Df    AIC
## - BPMEDS  1 660.50
## <none>      661.41
##
## Step:  AIC=660.5
## surv_male ~ SYSBP + AGE + DIABETES
##
##           Df    AIC
## <none>      660.50
## + BPMEDS  1 661.41

```

```
summary(step_male)
```

```

## Call:
## coxph(formula = surv_male ~ SYSBP + AGE + DIABETES, data = base_male)
##
##      n= 1908, number of events= 48
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## SYSBP      0.031160  1.031650 0.005502  5.664 1.48e-08 ***
## AGE        0.058057  1.059776 0.018733  3.099  0.00194 **
## DIABETES  1.297516  3.660195 0.418733  3.099  0.00194 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## SYSBP      1.032      0.9693      1.021      1.043
## AGE        1.060      0.9436      1.022      1.099
## DIABETES   3.660      0.2732      1.611      8.316

```

```
##
## Concordance= 0.789 (se = 0.034 )
## Likelihood ratio test= 63.33 on 3 df, p=1e-13
## Wald test = 79.71 on 3 df, p=<2e-16
## Score (logrank) test = 92.85 on 3 df, p=<2e-16

# female
# Female full model
full_female <- coxph(surv_female ~ SYSBP + AGE + DIABETES + BPMEDS,
                     data = base_female)

# Stepwise on female model
step_female <- stepAIC(full_female,
                       scope = list(lower = ~ SYSBP + AGE + DIABETES,
                                     upper = ~ SYSBP + AGE + DIABETES + BPMEDS),
                       direction = "both",
                       trace = TRUE)

## Start: AIC=838.58
## surv_female ~ SYSBP + AGE + DIABETES + BPMEDS
##
##           Df      AIC
## <none>      838.58
## - BPMEDS   1 838.59

summary(step_female)

## Call:
## coxph(formula = surv_female ~ SYSBP + AGE + DIABETES + BPMEDS,
##       data = base_female)
##
## n= 2434, number of events= 59
##
##           coef exp(coef) se(coef)      z Pr(>|z|)
## SYSBP      0.02497   1.02528  0.00425  5.874 4.24e-09 ***
## AGE        0.06609   1.06832  0.01918  3.445  0.00057 ***
## DIABETES   0.54124   1.71813  0.52055  1.040  0.29846
## BPMEDS     0.54668   1.72751  0.36746  1.488  0.13683
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##           exp(coef) exp(-coef) lower .95 upper .95
## SYSBP           1.025     0.9753    1.0168    1.034
## AGE             1.068     0.9360    1.0289    1.109
## DIABETES        1.718     0.5820    0.6194    4.766
## BPMEDS          1.728     0.5789    0.8407    3.550
##
## Concordance= 0.801 (se = 0.029 )
## Likelihood ratio test= 83.88 on 4 df, p=<2e-16
## Wald test = 92.02 on 4 df, p=<2e-16
## Score (logrank) test = 113.2 on 4 df, p=<2e-16
```

```

## remove bpmeds as not significant in males, barely in females
# add back in cursmoke because we need one extra profile

final_male <- coxph(surv_male ~ SYSBP + AGE +
                    DIABETES + CURSMOKE, data = base_male)
final_female <- coxph(surv_female ~ SYSBP + AGE +
                     DIABETES + CURSMOKE, data = base_female)

### check assumptions

# Schoenfeld residuals
ph_male <- cox.zph(final_male)
ph_female <- cox.zph(final_female)

print(ph_male)

```

```

##          chisq df    p
## SYSBP    2.399  1 0.121
## AGE      0.593  1 0.441
## DIABETES 3.717  1 0.054
## CURSMOKE 0.167  1 0.683
## GLOBAL   5.516  4 0.238

```

```

print(ph_female)

```

```

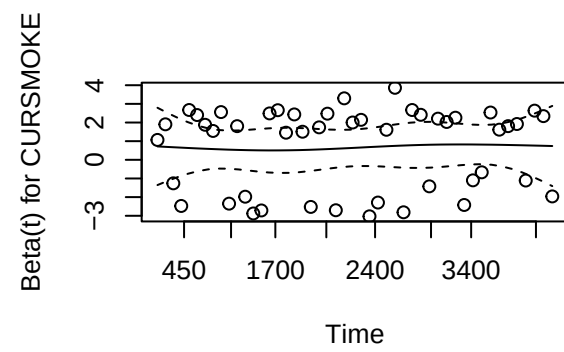
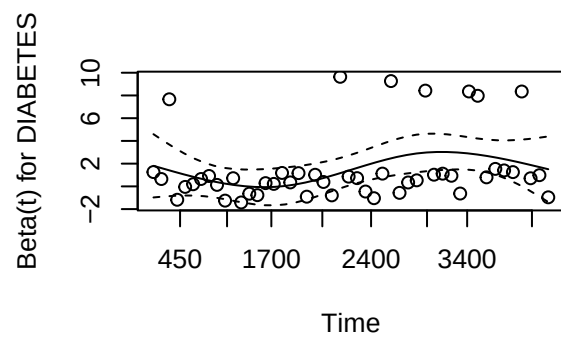
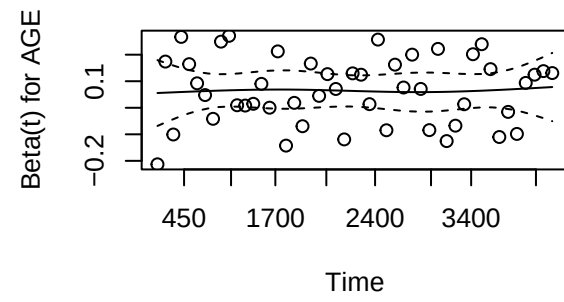
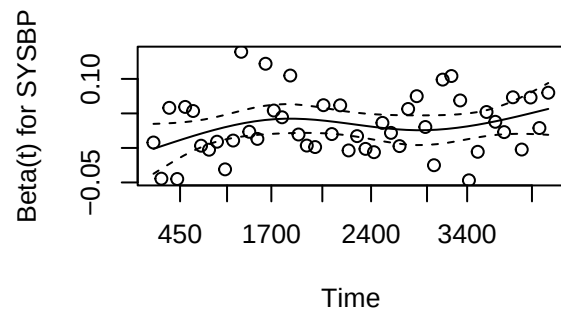
##          chisq df    p
## SYSBP    0.00388  1 0.95
## AGE      0.03515  1 0.85
## DIABETES 0.17667  1 0.67
## CURSMOKE 0.20685  1 0.65
## GLOBAL   0.43919  4 0.98

```

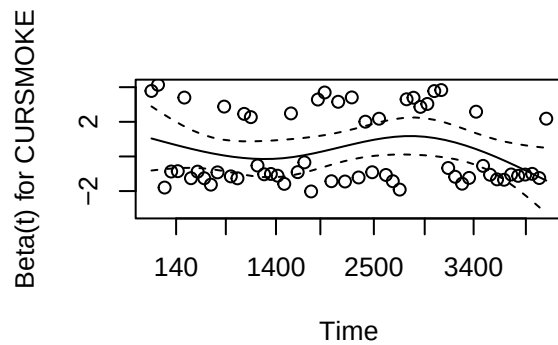
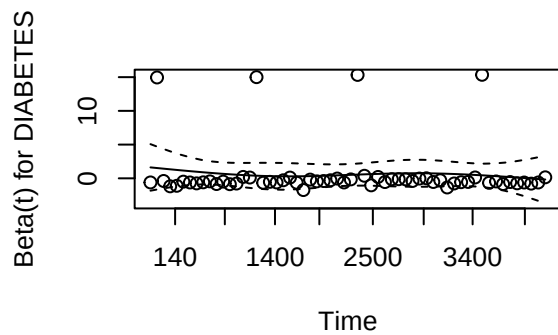
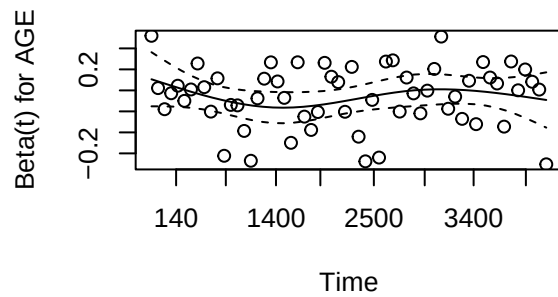
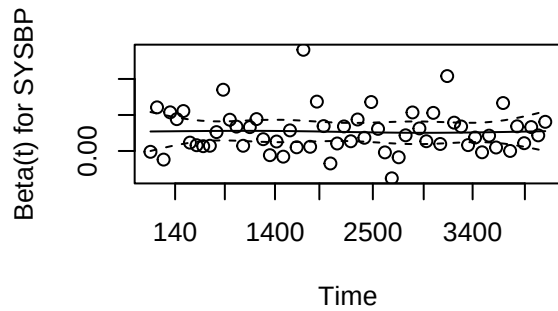
```

# Male
par(mfrow = c(2, 2))
plot(ph_male)

```



```
# Female
par(mfrow = c(2, 2))
plot(ph_female)
```



```
#### Survival Models ####
```

```
## profiles with age
```

```
#40
```

```
profiles_male_40 <- data.frame(
  SYSBP = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
            mean(base_male$SYSBP, na.rm = TRUE), 160,
            mean(base_male$SYSBP, na.rm = TRUE)),
  AGE = 40,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)
```

```
profiles_female_40 <- data.frame(
  SYSBP = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
            mean(base_female$SYSBP, na.rm = TRUE), 160,
            mean(base_female$SYSBP, na.rm = TRUE)),
  AGE = 40,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)
```

```
# Age 40
```

```
male_40 <- 1 - as.vector(summary(survfit(final_male, newdata = profiles_male_40),
                                     times = 3652.5)$surv)
```

```

female_40 <- 1 - as.vector(summary(survfit(final_female, newdata = profiles_female_40),
                                     times = 3652.5)$surv)

table_40 <- data.frame(
  Age      = 40,
  Profile  = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male     = round(1 - male_40, 4),
  Female   = round(1 - female_40, 4)
)

# 50
profiles_male_50 <- data.frame(
  SYSBP    = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE)),
  AGE      = 50,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

profiles_female_50 <- data.frame(
  SYSBP    = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
               mean(base_female$SYSBP, na.rm = TRUE), 160,
               mean(base_female$SYSBP, na.rm = TRUE)),
  AGE      = 50,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

# Age 50
male_50 <- 1 - as.vector(summary(survfit(final_male, newdata = profiles_male_50),
                                times = 3652.5)$surv)
female_50 <- 1 - as.vector(summary(survfit(final_female, newdata = profiles_female_50),
                                   times = 3652.5)$surv)

table_50 <- data.frame(
  Age      = 50,
  Profile  = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male     = round(1 - male_50, 4),
  Female   = round(1 - female_50, 4)
)

# 60
profiles_male_60 <- data.frame(
  SYSBP    = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE)),
  AGE      = 60,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

```

```

)

profiles_female_60 <- data.frame(
  SYSBP      = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
                  mean(base_female$SYSBP, na.rm = TRUE), 160,
                  mean(base_female$SYSBP, na.rm = TRUE)),
  AGE        = 60,
  DIABETES   = c(0, 0, 1, 1, 0),
  CURSMOKE   = c(0, 0, 0, 0, 1)
)

# Age 60
male_60      <- 1 - as.vector(summary(survfit(final_male, newdata = profiles_male_60),
                                     times = 3652.5)$surv)
female_60    <- 1 - as.vector(summary(survfit(final_female, newdata = profiles_female_60),
                                     times = 3652.5)$surv)

table_60 <- data.frame(
  Age        = 60,
  Profile     = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male        = round(1 - male_60, 4),
  Female      = round(1 - female_60, 4)
)

##### DO NOT USE THIS TABLE #####
# Final table for all of them

final_table <- rbind(table_40, table_50, table_60)
rownames(final_table) <- NULL

print(final_table)

```

```

##      Age      Profile   Male Female
## 1    40      Average 0.9944 0.9940
## 2    40      High BP 0.9861 0.9879
## 3    40    Diabetes 0.9753 0.9893
## 4    40 High BP + DM 0.9398 0.9784
## 5    40      Smoker 0.9890 0.9916
## 6    50      Average 0.9893 0.9878
## 7    50      High BP 0.9736 0.9755
## 8    50    Diabetes 0.9534 0.9783
## 9    50 High BP + DM 0.8883 0.9564
## 10   50      Smoker 0.9792 0.9830
## 11   60      Average 0.9797 0.9753
## 12   60      High BP 0.9503 0.9506
## 13   60    Diabetes 0.9131 0.9561
## 14   60 High BP + DM 0.7978 0.9131
## 15   60      Smoker 0.9606 0.9656

```

```
#####
```

```

# Try a different approach
# Age 40
m40 <- summary(survfit(final_male, newdata = profiles_male_40),
               times = 3652.5)$surv
f40 <- summary(survfit(final_female, newdata = profiles_female_40),
               times = 3652.5)$surv

# Age 50
m50 <- summary(survfit(final_male, newdata = profiles_male_50),
               times = 3652.5)$surv
f50 <- summary(survfit(final_female, newdata = profiles_female_50),
               times = 3652.5)$surv

# Age 60
m60 <- summary(survfit(final_male, newdata = profiles_male_60),
               times = 3652.5)$surv
f60 <- summary(survfit(final_female, newdata = profiles_female_60),
               times = 3652.5)$surv

# Build table manually indexing each profile [1,i]
final_table <- data.frame(
  Age      = rep(c(40, 50, 60), each = 5),
  Profile  = rep(c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"), 3),
  Male     = round(1 - c(m40[1,1], m40[1,2], m40[1,3], m40[1,4], m40[1,5],
                        m50[1,1], m50[1,2], m50[1,3], m50[1,4], m50[1,5],
                        m60[1,1], m60[1,2], m60[1,3], m60[1,4], m60[1,5]), 4),
  Female   = round(1 - c(f40[1,1], f40[1,2], f40[1,3], f40[1,4], f40[1,5],
                        f50[1,1], f50[1,2], f50[1,3], f50[1,4], f50[1,5],
                        f60[1,1], f60[1,2], f60[1,3], f60[1,4], f60[1,5]), 4)
)

print(final_table)

```

```

##   Age      Profile  Male Female
## 1  40      Average 0.0056 0.0060
## 2  40      High BP 0.0139 0.0121
## 3  40      Diabetes 0.0247 0.0107
## 4  40 High BP + DM 0.0602 0.0216
## 5  40      Smoker 0.0110 0.0084
## 6  50      Average 0.0107 0.0122
## 7  50      High BP 0.0264 0.0245
## 8  50      Diabetes 0.0466 0.0217
## 9  50 High BP + DM 0.1117 0.0436
## 10 50      Smoker 0.0208 0.0170
## 11 60      Average 0.0203 0.0247
## 12 60      High BP 0.0497 0.0494
## 13 60      Diabetes 0.0869 0.0439
## 14 60 High BP + DM 0.2022 0.0869
## 15 60      Smoker 0.0394 0.0344

```

```

# Option 1 - knitr alone (likely already installed)
library(knitr)
kable(final_table,

```

```
col.names = c("Age", "Risk Profile", "Male", "Female"),
caption = "10-Year Stroke Probability by Risk Profile, Age, and Sex",
align = c("c", "l", "c", "c"))
```

Table 1: 10-Year Stroke Probability by Risk Profile, Age, and Sex

Age	Risk Profile	Male	Female
40	Average	0.0056	0.0060
40	High BP	0.0139	0.0121
40	Diabetes	0.0247	0.0107
40	High BP + DM	0.0602	0.0216
40	Smoker	0.0110	0.0084
50	Average	0.0107	0.0122
50	High BP	0.0264	0.0245
50	Diabetes	0.0466	0.0217
50	High BP + DM	0.1117	0.0436
50	Smoker	0.0208	0.0170
60	Average	0.0203	0.0247
60	High BP	0.0497	0.0494
60	Diabetes	0.0869	0.0439
60	High BP + DM	0.2022	0.0869
60	Smoker	0.0394	0.0344